

W04 · M04 · 15 MINIT / MINUTES · Hari 1, 16:40 hingga 16:55

Hands-On SQA-AI Assistant Workshop

AI test design: EP, BVA and coverage gaps

Reka bentuk ujian AI: EP, BVA dan jurang liputan

Generate D13-format test cases with named techniques for two checkout requirements, then ask the AI to find RTM coverage gaps.

opencode 1.18 · model Big Pickle (OpenCode Zen, free, no API key) · synthetic ShopFast data only

Bengkel SQA KRISAv2 dan PPrISA 2.0 · 14 hingga 15 September 2026 · Makmal Komputer Aras 11, Menara MyIPO

[Web guide \(copy prompts\)](#)

[Module hub](#)

[Example outputs \(GitHub\)](#)

Before you start / Sebelum mula

YOU WILL / ANDA AKAN

1. Run /testcases for REQ-CHK-01 and REQ-CHK-03
2. Approve the CSV and check boundaries 0, 1, 10, 11 and RM200.00
3. Ask the Plan agent for RTM coverage gaps
4. Merge the good cases into your Lab B sheet

1. Jalankan /testcases untuk REQ-CHK-01 dan REQ-CHK-03
2. Luluskan CSV dan semak sempadan 0, 1, 10, 11 dan RM200.00
3. Minta ejen Plan mencari jurang liputan RTM
4. Gabungkan kes yang baik ke dalam helaian Makmal B

CHECK FIRST / SEMAK DAHULU

- PowerShell is in C:\Users\<you>\sqa-krisa-bengkel\lab
- opencode starts and shows Big Pickle, OpenCode Zen (W00)
- Outputs from earlier workshops exist in outputs\ (or copy them from example-output)

Data rules / Peraturan data

- Synthetic ShopFast data only. Never paste real agency documents or personal data.
- Free models run outside Malaysia and may learn from prompts.
- AI output is a draft. A person checks and signs off.

OUTPUT

outputs/m04-testcases.csv

W04 · STEP 1 OF 4 · LANGKAH 1

Run /testcases for two checkout requirements. The AI reads the documents first and may pause at Preparing write. When the Permission required dialog appears, read the preview and choose Allow once.

Jalankan /testcases untuk dua keperluan checkout. AI membaca dokumen dahulu dan mungkin berhenti seketika pada Preparing write. Apabila dialog Permission required muncul, baca pratonton dan pilih Allow once.

TYPE IN OPENCODE (BUILD)

/testcases REQ-CHK-01 dan REQ-CHK-03, maksimum 10 kes ujian

OC | Mewujudkan Kes Ujian untuk REQ-CHK-01...

Peraturan	Status ahli	Kod promosi	Subtotal	Diskaun ahli	Diskaun promosi	Diskaun	Penghantaran
A: Ahli + RAYA15, sub=200	Ya	RAYA15	200.00	20.00	30.00	50.00	0
B: Ahli + MEGA20, sub=250	Ya	MEGA20	250.00	25.00	50.00	75.00	0
C: Ahli + tiada kod, sub=100	Ya	Tiada	100.00	10.00	0	10.00	8.00
D: Ahli + RAYA15, sub=199	Ya	RAYA15	199.00	19.90	29.85	49.75	8.00
E: Bukan ahli + tiada kod, sub=179.80	Tidak	Tiada	179.80	0	0	0	8.00
F: Ahli + had 25% cap, sub=200	Ya	MEGA20	200.00	20.00	30.00	<= 50.00	0

Sekarang saya tulis fail CSV dan paparkan ringkasan.

Write outputs\m04-testcases.csv

Build · Big Pickle

△ Permission required

→ Edit outputs\m04-testcases.csv

1 + TC_ID,FR_ID,Tajuk,Teknik,Partisi_Atau_Peraturan,Prasyarat,Langkah>Data_Ujian,Jangkaan,Keutamaan,Keputusan

2 + TC-CHK-001,REQ-CHK-001,Kuantiti 2 diterima,Sah,BVA sempadan minimum: kuantiti 1,Sesi troli baharu tanpa item,1. POST /api/cart/items dengan productId P001 dan kuantiti 1; 2. Semak kod

Allow once

Allow always

Reject

ctrl+f fullscreen

select

enter confirm

Mewujudkan Kes Ujian untuk REQ-CHK-01 dan REQ-CHK-03

Context
31,394 tokens
16% used
\$0.00 spent

LSP
LSPs are disabled

Getting started

OpenCode includes free models so you can start immediately.

Connect from 75+ providers to use other models, including Claude, GPT, Gemini etc

Connect provider /connect

/~\trainee-demo\sqa-krisa-bengkel\lab: main

• OpenCode 1.18.30

Allow once. The D13-format test cases are saved.

Allow once. Kes ujian format D13 disimpan.

TYPE IN OPENCODE

Allow once (Enter)

OC | Mewujudkan Kes Ujian untuk REQ-CHK-01...

Teknik	Bilangan Kes	TC_ID
BVA	4	TC-CHK-001, TC-CHK-002, TC-CHK-003, TC-CHK-004
EP	2	TC-CHK-005, TC-CHK-006
BVA + Decision Table	1	TC-CHK-007
Decision Table	3	TC-CHK-008, TC-CHK-009, TC-CHK-010
Jumlah	10	TC-CHK-001 hingga TC-CHK-010

Rujukan

- KRISAv2 Bab 6, 6.7 Dokumentasi Persediaan Ujian (../references/krisa-v2-beta-2026/BAB6-FASA-PENGUJIAN-PENERIMAAN.pdf#page=9)
- KRISAv2 Bab 5, 5.7 Pengujian Sistem (../references/krisa-v2-beta-2026/BAB5-FASA-PEMBANGUNAN.pdf#page=33)
- Keperluan sumber: BAB3-FASA-ANALISIS.pdf#page=42 (../references/krisa-v2-beta-2026/BAB3-FASA-ANALISIS.pdf#page=42) (Spesifikasi Keperluan Sistem, F2.6)

Nota: Peraturan B dan D (senario subtotal RM250 MEGA20 dan subtotal RM199.99) tidak dimasukkan kerana had 10 kes bagi semakan ini. TC-CHK-010 menguji had cap 25% secara direktif dengan had toleransi diskaun kerana SRS tidak menyatakan algoritma peruntukan apabila gabungan melebihi 25%.

Semak manusia / Human check

1. Setiap parti dan nilai sempadan yang disenaraikan mempunyai kes ujian, termasuk kedua-dua belah setiap sempadan.
2. Setiap Jangkaan sepadan dengan kriteria penerimaan SRS atau openapi.yaml, bukan dengan tingkah laku aplikasi yang diperhatikan.
3. Kes ujian boleh dilaksanakan oleh rakan lain tanpa bertanya: prasyarat, data dan langkah lengkap.

Build · Big Pickle · 4m 6s

Build · Big Pickle OpenCode Zen

C:\Users\user\trainee-demo\sqa-krisa-bengkel\lab38.4K (19%)ctrl+p commands

Mewujudkan Kes Ujian untuk REQ-CHK-01 dan REQ-CHK-03

Context
38,368 tokens
19% used
\$0.00 spent

LSP
LSPs are disabled

Getting started
OpenCode includes free models so you can start immediately.
Connect from 75+ providers to use other models, including Claude, GPT, Gemini etc
Connect provider /connect

/~\trainee-demo\sqa-krisa-bengkel\lab: main

• OpenCode 1.18.30

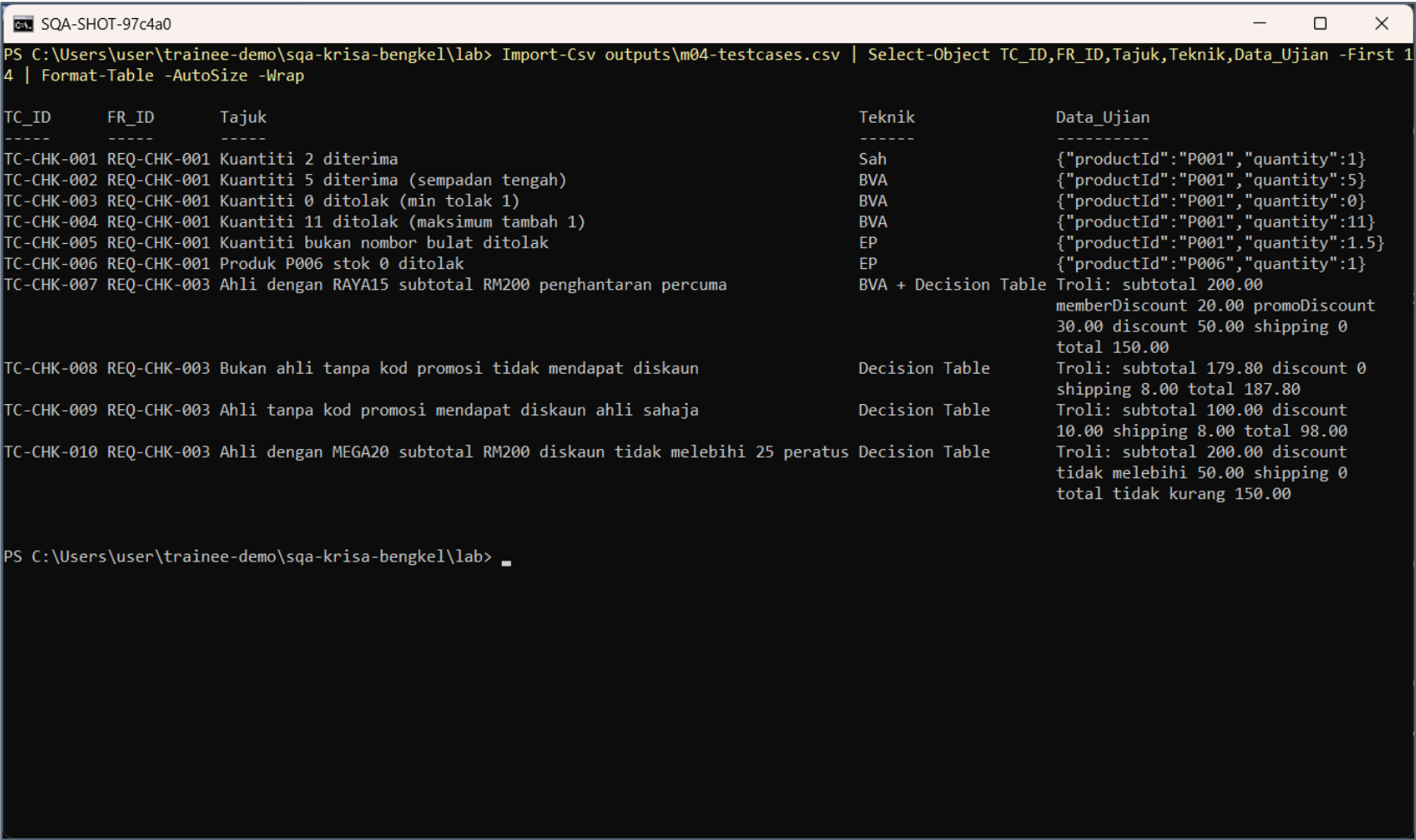
W04 · STEP 3 OF 4 · LANGKAH 3

Open the CSV and check it. In this real run the AI wrote REQ-CHK-001 instead of REQ-CHK-01, labelled one case Sah instead of a technique, and missed boundary 10, the value most likely to hide a bug. Fix these before using the file.

Buka CSV dan semak. Dalam larian sebenar ini AI menulis REQ-CHK-001 dan bukan REQ-CHK-01, melabel satu kes Sah dan bukan teknik, dan terlepas sempadan 10, nilai yang paling mungkin menyembunyikan pepijat. Betulkan sebelum fail digunakan.

TYPE IN POWERSHELL

```
Import-Csv outputs\m04-testcases.csv | Format-Table -AutoSize -Wrap
```



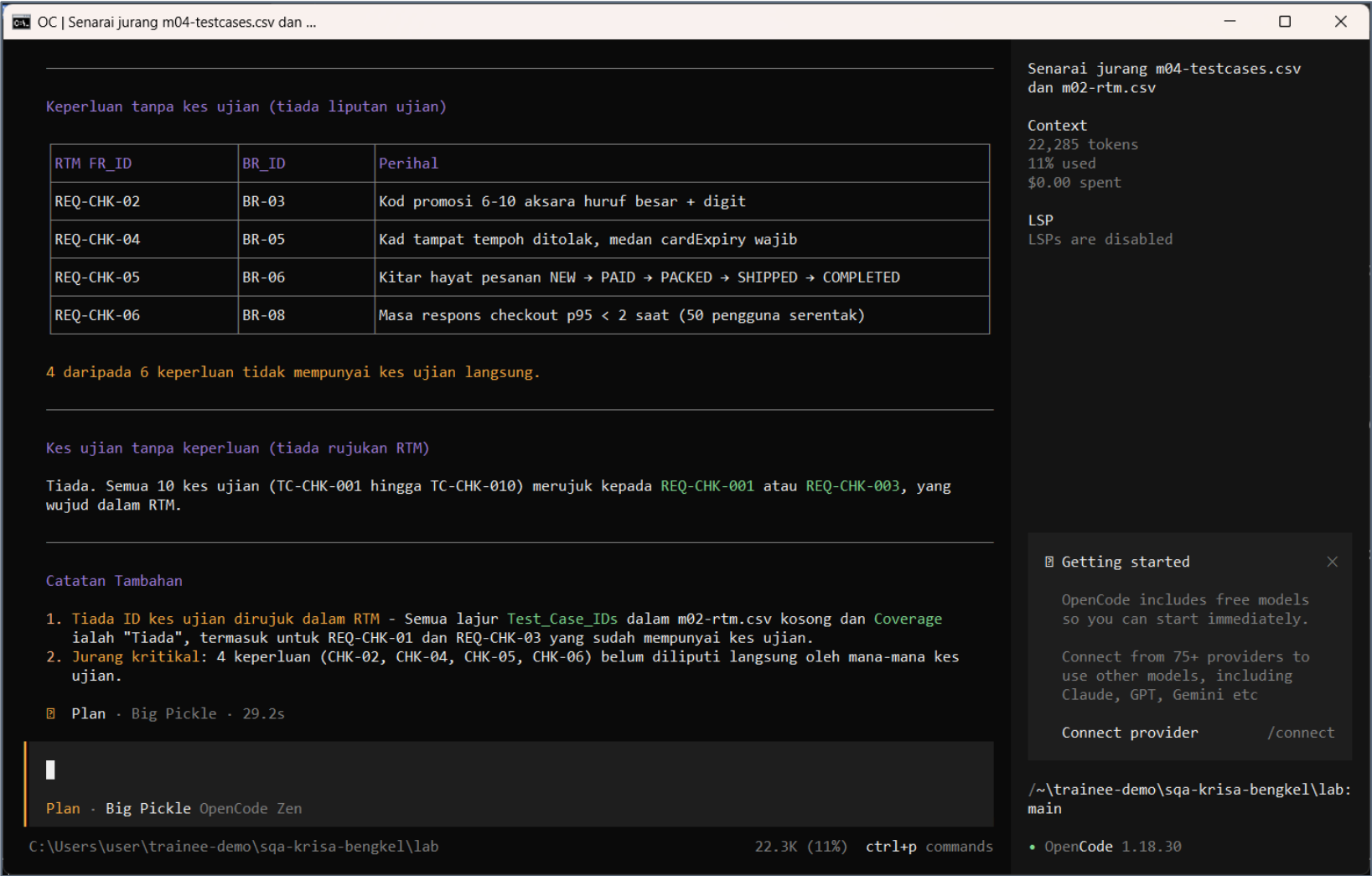
W04 · STEP 4 OF 4 · LANGKAH 4

Ask the Plan agent to compare test cases with the RTM. It correctly lists 4 uncovered requirements, but claims REQ-CHK-001 exists in the RTM when the RTM uses REQ-CHK-01. The AI does not check its own IDs: you must.

Minta ejen Plan membandingkan kes ujian dengan RTM. Ia betul menyenaraikan 4 keperluan tanpa liputan, tetapi mendakwa REQ-CHK-001 wujud dalam RTM sedangkan RTM menggunakan REQ-CHK-01. AI tidak menyemak ID sendiri: anda yang mesti menyemak.

TYPE IN OPENCODE (PLAN)

Bandingkan outputs/m04-testcases.csv dengan outputs/m02-rtm.csv. Senaraikan keperluan tanpa kes ujian dan kes ujian tanpa keperluan. Jangan tulis fail.



Human check before you accept the output / Semakan manusia

Boundary values 10 and RM200.00 are present

Nilai sempadan 10 dan RM200.00 ada

Every case names its technique and REQ-CHK ID

Setiap kes menamakan teknik dan ID REQ-CHK

Expected results come from the SRS, not from what the app does

Hasil dijangka datang daripada SRS, bukan daripada tingkah laku aplikasi

The AI draft becomes evidence only after this check. Record who checked it and when (Module 7 audit trail).

If something goes wrong / Jika ada masalah

opencode is not recognised

Close and reopen PowerShell so PATH refreshes. Still failing: `powershell -ExecutionPolicy Bypass -File .\setup\install-opencode.ps1 -Portable`

Scripts are disabled on this system

`Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`, or run `opencode.cmd` instead of `opencode`

The model is slow or stops

Wait up to 2 minutes. Press Esc to interrupt, then send the prompt again. Try another free model with `/models`

Permission required dialog

Read the preview. Choose Allow once for files in `outputs/`. Reject anything outside the lab folder

File or command not found

Check the status bar path ends with `\lab`. Commands and `@files` are relative to the lab folder

Answer uses Indonesian words or dashes

Reply: Ikut AGENTS.md peraturan 1 dan 14, tulis semula dalam Bahasa Melayu Malaysia

Network or proxy error

`$env:HTTPS_PROXY = "http://proxy:port"` and `$env:NO_PROXY = "localhost,127.0.0.1"` before starting opencode

Port 3000 already in use

`$env:PORT = 3001` before npm start, and use `http://localhost:3001`

Abbreviations / Singkatan (1 of 3)

Short forms used in this workshop, in plain words. / Singkatan dalam bengkel ini, dengan bahasa mudah.

Abbreviation / Singkatan	Stands for / Maksud penuh	In plain words / Maksud mudah
AI	Artificial Intelligence	Software that can answer questions and write text or code by learning patterns from large amounts of data. <i>Kecerdasan Buatan. Perisian yang boleh menjawab soalan dan menulis teks atau kod berdasarkan pola daripada data yang banyak.</i>
API	Application Programming Interface	A doorway that lets one program ask another program for data or an action, without using a screen. <i>Pintu yang membolehkan satu program meminta data atau tindakan daripada program lain tanpa melalui skrin.</i>
BVA	Boundary Value Analysis	Testing at the edges of the allowed values, such as 0, 1, 10 and 11, because mistakes often hide there. <i>Analisis Nilai Sempadan. Menguji di hujung julat nilai seperti 0, 1, 10 dan 11 kerana kesilapan kerap berlaku di situ.</i>
CSV	Comma-Separated Values	A simple table file that opens in Excel, with commas between the columns. <i>Fail jadual ringkas yang boleh dibuka dalam Excel, dengan koma di antara lajur.</i>
EP	Equivalence Partitioning	Grouping inputs that behave the same way, then testing one value from each group. <i>Pembahagian Kesetaraan. Mengumpulkan input yang berkelakuan sama, kemudian menguji satu nilai daripada setiap kumpulan.</i>
FAT	Factory Acceptance Test	The formal test with the customer before the system is installed at the live site. <i>Ujian Penerimaan Kilang. Ujian formal bersama pelanggan sebelum sistem dipasang di tapak sebenar.</i>

Abbreviations / Singkatan (2 of 3)

Short forms used in this workshop, in plain words. / Singkatan dalam bengkel ini, dengan bahasa mudah.

Abbreviation / Singkatan	Stands for / Maksud penuh	In plain words / Maksud mudah
ID	Identifier	A unique label, such as REQ-CHK-01, so that each item can be traced. <i>Pengecam, iaitu label unik seperti REQ-CHK-01 supaya setiap item boleh dijejak.</i>
KRISA	Kejuruteraan Sistem Aplikasi Sektor Awam	The Malaysian public sector guide for engineering application systems. KRISAv2 is the 2026 beta edition. <i>Buku Panduan Kejuruteraan Sistem Aplikasi Sektor Awam. KRISAv2 ialah edisi beta 2026.</i>
MyIPO	Perbadanan Harta Intelek Malaysia	Intellectual Property Corporation of Malaysia, the host of this training venue. <i>Perbadanan Harta Intelek Malaysia, tuan rumah lokasi latihan ini.</i>
PAT	Provisional Acceptance Test	The KRISA acceptance test done before provisional acceptance of the system, including performance checks. <i>Ujian Penerimaan Sementara dalam KRISA sebelum sistem diterima secara sementara, termasuk semakan prestasi.</i>
PPrISA	Panduan Pengurusan Projek ICT Sektor Awam	The Malaysian public sector guide for managing ICT projects. Codes such as PPrISA05 are its templates. <i>Panduan Pengurusan Projek ICT Sektor Awam. Kod seperti PPrISA05 ialah templatnya.</i>
REQ	Requirement	Prefix in requirement IDs such as REQ-CHK-01. CHK stands for checkout. <i>Keperluan. Awalan dalam ID keperluan seperti REQ-CHK-01. CHK bermaksud checkout.</i>

Abbreviations / Singkatan (3 of 3)

Short forms used in this workshop, in plain words. / Singkatan dalam bengkel ini, dengan bahasa mudah.

Abbreviation / Singkatan	Stands for / Maksud penuh	In plain words / Maksud mudah
RM	Ringgit Malaysia	Malaysian currency. <i>Mata wang Malaysia.</i>
RTM	Requirements Traceability Matrix	A table linking each requirement to its design, tests and defects, so nothing is missed. <i>Matriks Kebolehkesanan Keperluan. Jadual yang memautkan setiap keperluan kepada reka bentuk, ujian dan kecacatan supaya tiada yang tertinggal.</i>
SQA	Software Quality Assurance	All the planned work that gives confidence the software meets its quality requirements. <i>Jaminan Kualiti Perisian, iaitu semua kerja terancang yang memberi keyakinan bahawa perisian memenuhi keperluan kualiti.</i>
SRS	Spesifikasi Keperluan Sistem	The System (or Software) Requirements Specification: what the system must do. KRISA deliverable D03. <i>Spesifikasi Keperluan Sistem, iaitu perkara yang mesti dilakukan oleh sistem. Serahan KRISA D03.</i>
TUI	Text User Interface	A full-screen program that runs inside a terminal window, like opencode. <i>Antara muka pengguna berasaskan teks, iaitu program skrin penuh dalam tettingkap terminal seperti opencode.</i>
UAT	User Acceptance Test	Users try the system to confirm it meets their needs before they accept it. <i>Ujian Penerimaan Pengguna. Pengguna mencuba sistem untuk memastikan ia memenuhi keperluan sebelum menerimanya.</i>

Links / Pautan

[Web guide with copyable prompts](#)

[Module hub \(slides, references, templates\)](#)

[Example outputs from the real run](#)

[Lab README and command list](#)

[AGENTS.md rules used by the assistant](#)

[opencode documentation](#)

[opencode TUI commands and keys](#)

[OpenCode Zen free models and data policy](#)

[Bab 5, 5.7 Pengujian Sistem, peringkat ujian \(p.33\)](#)

[Bab 6, 6.6 Pelan Induk Pengujian F5.1 \(p.4\)](#)

[Bab 6, Jadual 6.2 Peringkat UAT, PAT, FAT \(p.5\)](#)